



AI & Partners

Amsterdam - London - Singapore



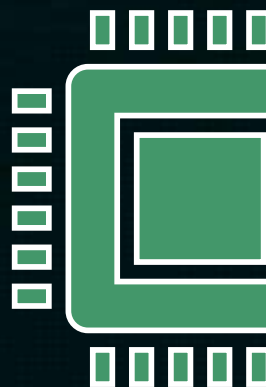
EU AI Act

Serious Incident Reporting:
Ensuring Public Safety

October 2025

Authors

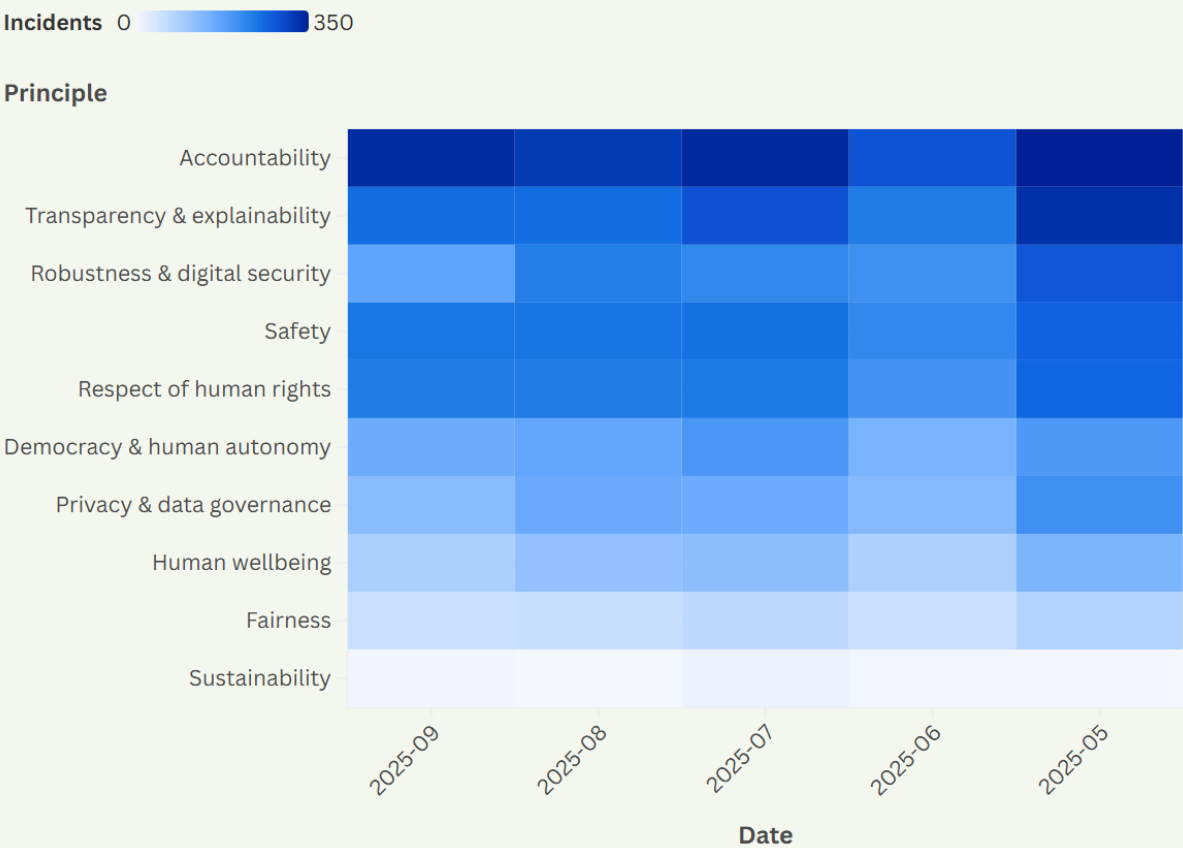
Sean Musch, Michael Borrelli, Charles Kerrigan, Félicien Vallet,
Shraeyas Jayasimha R, Juhi Koré, Jeremy Prasetyo





Accountability and Transparency Dominate Emerging AI Incident Trends, While Ethical and Sustainability Risks Remain Underreported

OECD AI: Evolution of Incidents and Hazards by AI Principle (May – September 2025)



The forward-looking data indicates evolving priorities in AI incident monitoring, with accountability remaining the most frequently reported area (336 in September 2025). Transparency, safety, and robustness also show high levels of attention, though their recent declines may suggest growing system resilience or a narrowing of reporting criteria.

In contrast, consistently low figures for fairness, human wellbeing, and sustainability point to potential blind spots where risks may be under-recognised or under-measured. These areas may require clearer definitions and stronger reporting mechanisms to ensure a holistic approach to AI risk management.

Although distinct from formal reporting under Article 73 of the EU AI Act, these patterns offer insight into where serious incidents *could* emerge. They highlight the importance of building balanced, proactive reporting frameworks that extend beyond governance and safety toward social, ethical, and environmental dimensions of AI impact.



AI & Partners

Amsterdam - London - Singapore

AI & Partners defends and extends the digital rights of users at risk around the world. By combining direct technical support, comprehensive policy engagement, global advocacy, grassroots professional services, regulatory interventions, and participating in industry groups such as AI Commons, we fight for fundamental rights in the artificial intelligence age.

This report was prepared by Sean Donald John Musch and Michael Charles Borrelli. For more information visit <https://www.ai-and-partners.com/>.

Contact: contact@ai-and-partners.com.

This report is an AI & Partners publication.



AI & Partners
Amsterdam - London - Singapore



AI & Partners

Amsterdam - London - Singapore

Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We've seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it's handled responsibly. That's why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems
are unknown

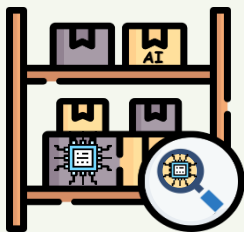
What Do We Do?

We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.

How Do We Do It?

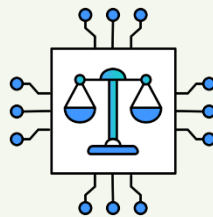
Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building 'Trustworthy-by-Design'.



Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.





Contents

Who Are We	3
Introduction	5
Key questions being asked about Serious Incident Reporting	6
1. What is a serious incident in the context of AI systems?	7
2. Why is it important to report serious incidents related to AI systems?	7
3. How soon must a serious incident be reported?	7
4. What should be included in a serious incident report?	7
5. What happens after a serious incident report is submitted?	7
6. Who is responsible for reporting serious incidents?	8
7. Can a serious incident report be submitted if the cause is uncertain?	8
8. What corrective actions must be taken after a serious incident?	8
9. Are there any exceptions to the serious incident reporting requirement?	8
10. What role do market surveillance authorities play in serious incident reporting?	8
Understanding Serious Incident Reporting	9
1. Physical	10
2. Psychological	11
3. Economic/Property	12
4. Reputational	13
5. Public Interest	14
6. Human Rights	15
Mapping Conformity Assessments to ecosystem actors	16
Calls to Action	0
Conclusion	30
References	31





Introduction

As the EU Artificial Intelligence Act (EU AI Act) advances towards full implementation, the reporting of serious incidents under Article 73 emerges as a critical mechanism for maintaining the safety, accountability, and fundamental rights protections envisioned by the legislation. Article 73 requires providers and, where relevant, deployers of high-risk AI systems to notify competent authorities of any serious incidents or malfunctioning that pose risks to health, safety, or fundamental rights. This obligation forms part of the EU's broader effort to ensure that post-market monitoring is not merely reactive, but a proactive and transparent process aligned with public interest.

This paper provides a practical framework for stakeholders navigating the requirements and operational challenges associated with serious incident reporting under Article 73. It draws on lessons from parallel regulatory fields—such as medical devices, cybersecurity, and data protection—to extract best practices for identifying, documenting, and communicating serious incidents in a timely and structured manner. The aim is to translate legal obligations into operational protocols that support compliance while reinforcing public trust in AI technologies.

Through analysis of real-world incident scenarios and reporting chains, the paper highlights the importance of integrating incident detection and escalation procedures throughout the AI lifecycle—from initial development through deployment and ongoing oversight. Emphasis is placed on the need for internal governance structures that enable clear accountability, ethical foresight, and legal preparedness, ensuring that serious incidents are not only reported in accordance with Article 73 but are also used as learning opportunities to improve system safety and fairness. In mapping the roles and responsibilities of key actors—including providers, deployers, authorised representatives, notified bodies, and regulators—the paper promotes a coherent and collaborative approach to incident reporting. It underscores how systematic reporting can serve not only as a compliance measure but also as a tool for enhancing risk management, informing regulatory oversight, and aligning AI systems with EU values such as transparency, non-discrimination, and human dignity.

Whether you are a technology provider seeking to meet post-market obligations, a regulator tasked with assessing incident trends, or a civil society organisation monitoring AI harms, this paper offers structured guidance for engaging with Article 73 reporting. It lays the groundwork for a responsive, resilient, and rights-respecting AI ecosystem across the EU Single Market.

Best regards,

Sean Musch

Founder/CEO

AI & Partners





Key questions being asked about Serious Incident Reporting



1. What is a serious incident in the context of AI systems?

A serious incident refers to an event in which an AI system causes significant harm, risk, or potential damage to individuals, organizations, or society. This can include physical harm, psychological distress, economic loss, or violation of human rights. A serious incident is considered when there is a direct or reasonable likelihood of causal connection between the AI system's performance and the incident. In AI deployment, such incidents can involve accidents with autonomous vehicles, biased decision-making, or privacy violations due to AI misuse, all of which must be reported to the relevant authorities.

2. Why is it important to report serious incidents related to AI systems?

Reporting serious incidents involving AI is crucial to ensuring public safety, consumer protection, and accountability. Timely reporting allows regulators to respond quickly, conduct investigations, and mitigate the risk of similar incidents occurring in the future. It helps identify systemic issues with AI algorithms, including bias, inaccuracies, or safety failures. Reporting fosters transparency, facilitates regulatory oversight, and encourages organizations to take corrective actions. By ensuring that incidents are promptly reported, stakeholders—such as consumers, workers, and businesses—can have confidence that AI systems are being held to high safety and ethical standards.

3. How soon must a serious incident be reported?

According to the regulation, AI system providers must report serious incidents to the relevant authorities as soon as they establish a causal link or reasonable suspicion of such a link between the AI system and the incident. Generally, the report must be submitted within 15 days. If the incident is widespread or involves severe harm, such as death or significant risk, the report must be made within two days. If a fatality occurs, the report must be submitted within 10 days. Initial incomplete reports may be submitted, followed by a complete report.

4. What should be included in a serious incident report?

A serious incident report should include details of the incident, including the nature and extent of the harm caused, the suspected cause of the incident, the steps taken to investigate the issue, and any immediate actions or mitigations implemented to prevent further harm. The report should also contain a risk assessment and an outline of corrective actions planned or already undertaken. Depending on the severity of the incident, companies may submit an initial incomplete report, which must be followed up with a full and comprehensive report once further investigations are complete.

5. What happens after a serious incident report is submitted?

Once a serious incident report is submitted, the market surveillance authority (MSA) or relevant regulatory body will assess the information. They will evaluate the extent of the harm and take appropriate actions, which may include initiating further investigations, issuing corrective orders, or even imposing fines or sanctions. In cases of widespread incidents or fatal consequences, the authorities may take immediate action to prevent further harm. Additionally, the provider of the AI system must cooperate with the investigation, including providing access to necessary data and information, and follow any corrective actions recommended by the authorities.





6. Who is responsible for reporting serious incidents?

The responsibility for reporting serious incidents lies with the provider of the AI system, which is the entity that developed, deployed, or placed the AI system on the market. In some cases, the deployer (i.e., the organization using the AI system in a specific context) may also have obligations depending on the nature of the incident and contractual arrangements. The provider must report incidents to the relevant market surveillance authority in the country where the incident occurred. Failure to report incidents may result in penalties, including fines and reputational damage.

7. Can a serious incident report be submitted if the cause is uncertain?

Yes, an initial serious incident report can be submitted even if the cause of the incident is not fully determined. If there is a reasonable suspicion or likelihood that the AI system is linked to the incident, the report must be made promptly—within the specified time frame (15 days, 2 days for widespread issues, or 10 days for fatalities). However, the provider must follow up with a complete report once the investigation is concluded and the cause is clearly identified. Transparency during this process is critical for building trust and addressing regulatory concerns.

8. What corrective actions must be taken after a serious incident?

After a serious incident, the provider of the AI system is required to take immediate and thorough corrective actions to prevent further harm. This includes performing a comprehensive risk assessment, investigating the root cause of the incident, and implementing necessary changes or updates to the system to address identified vulnerabilities. Providers must also cooperate with the regulatory authorities during the investigation, sharing relevant data and information. Depending on the findings, corrective actions may involve system modifications, updates to safety protocols, or adjustments to operational procedures to enhance overall safety and performance.

9. Are there any exceptions to the serious incident reporting requirement?

Yes, there are a few exceptions. For high-risk AI systems that are already governed by other EU legislation (e.g., medical devices or safety-critical equipment), the incident reporting may be subject to specific regulations under those laws. Additionally, if the AI system is a safety component within larger regulated devices, the reporting obligation might be limited to specific types of incidents as defined by the regulation. The scope of reporting obligations can vary depending on the type of AI system and its context, so it is important to understand how the regulations apply to specific use cases.

10. What role do market surveillance authorities play in serious incident reporting?

Market surveillance authorities (MSAs) are responsible for overseeing the compliance of AI systems with relevant safety and regulatory standards. When a serious incident report is submitted, the MSA reviews the information to assess the risk and severity of the incident. They may initiate investigations, request further data from the provider, and take corrective actions if necessary. In cases where there is a widespread impact or serious harm, MSAs have the authority to enforce regulations, issue sanctions, and mandate the recall or modification of AI systems. They play a key role in ensuring that AI systems are safe and reliable for users.

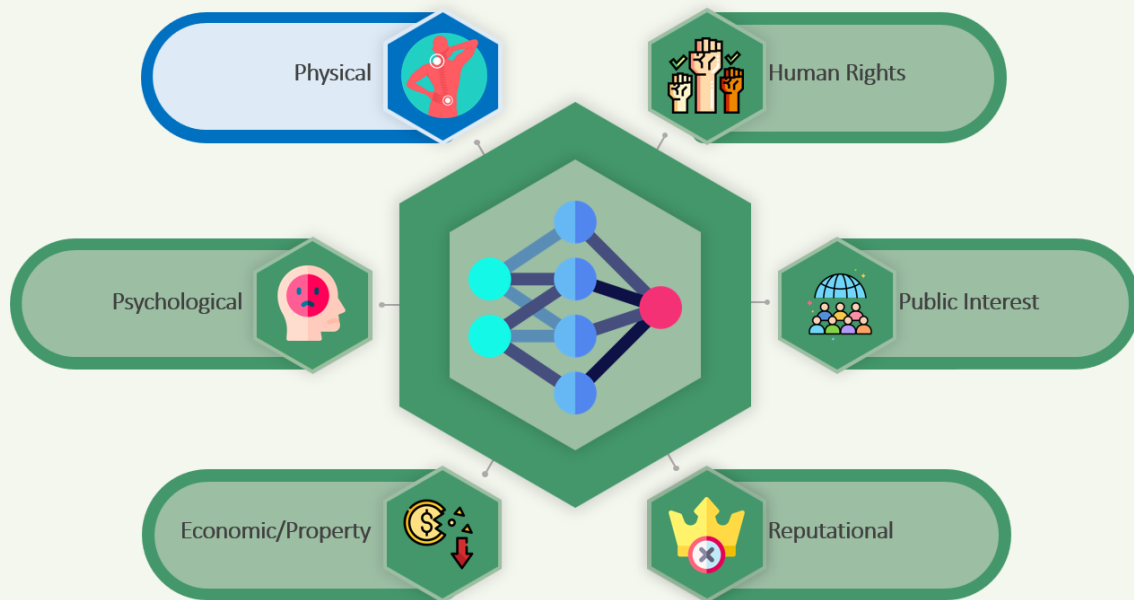




Understanding Serious Incident Reporting



1. Physical



What is this?

Physical harm refers to injury, illness, or death resulting from the malfunction or misuse of AI systems. This can include accidents caused by autonomous vehicles, faulty robotics in the workplace, or misdiagnoses by AI-driven medical devices. Such harm may result from programming errors, design flaws, inadequate testing, or real-time decision-making failures. The consequences are often immediate and visible, making physical harm one of the most pressing concerns in AI safety. Preventing physical harm requires stringent safety standards, real-time monitoring, transparent reporting processes, and systems that prioritize human safety in all contexts where AI interacts with people.

How is it relevant?

Physical harm is highly relevant to the deployment and regulation of AI systems, especially in sectors such as transportation, healthcare, manufacturing, and consumer electronics. As AI systems increasingly make autonomous decisions in real-world environments, the risk of physical accidents rises. Stakeholders including regulators, developers, and users must consider this risk in design, testing, and operational phases. Regulations like the EU AI Act mandate reporting of serious incidents to ensure accountability. Given AI's potential to operate unsupervised or respond to real-time stimuli, ensuring physical safety is critical to public acceptance and legal compliance.

Why does it matter?

Physical harm from AI can lead to loss of life, long-term injuries, and emotional trauma for victims and their families. These outcomes have far-reaching implications, including legal liability, financial costs, public distrust, and regulatory crackdowns. Incidents can stall technological progress, prompt mass recalls, or deter innovation due to fear of risk. Moreover, failure to address these harms disproportionately affects vulnerable populations such as the elderly, children, and persons with disabilities. Ensuring safety builds confidence in AI and supports ethical development. It reinforces the social contract that innovation should enhance human well-being rather than jeopardize it.





2. Psychological



What is this?

Psychological harm refers to emotional or mental distress caused by AI systems, including anxiety, stress, depression, fear, or diminished self-worth. It may arise from invasive surveillance, biased or opaque decision-making, or manipulative content delivery. For instance, algorithmic feeds that promote harmful content or job automation that fosters insecurity can contribute to mental health issues. Unlike physical harm, psychological harm is less visible but equally serious, affecting a person's cognitive, emotional, and social well-being. Addressing it requires ethical AI design, transparency, and mechanisms for human oversight and redress, particularly in sectors like employment, education, and social media.

How is it relevant?

Psychological harm is increasingly relevant in a digital society where AI shapes daily experiences, decisions, and social interactions. With AI systems involved in hiring, policing, education, and online platforms, the potential for emotional manipulation, unfair treatment, or social isolation has grown. Individuals subjected to opaque or biased AI judgments may feel powerless or dehumanized. Long-term exposure to algorithmically curated environments can reinforce negative beliefs, foster division, or exacerbate mental health crises. Therefore, psychological harm must be factored into regulatory assessments, ethical guidelines, and the design of AI systems intended for broad public use.

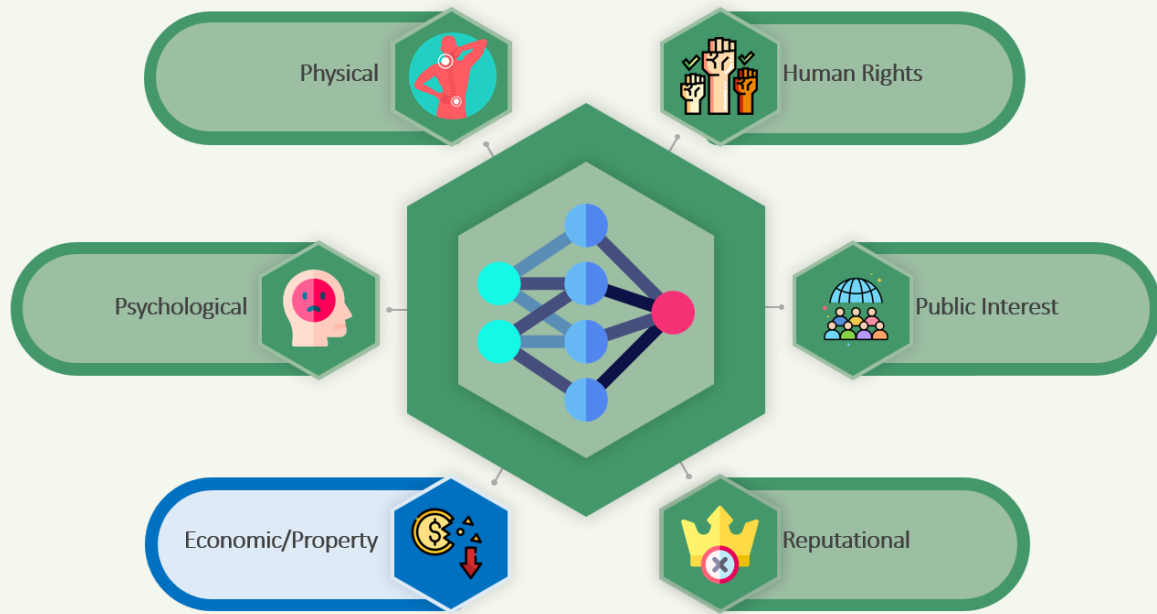
Why does it matter?

Psychological harm impacts individuals' mental health, self-esteem, and quality of life. At a broader level, it can erode societal trust in institutions and technology, leading to public disengagement or unrest. Mental health challenges also carry economic and social costs, including healthcare burdens, decreased productivity, and weakened community cohesion. For vulnerable populations—such as minorities, children, or persons with disabilities—psychological harms may deepen marginalization. Ethical AI must consider not only functionality but also emotional and cognitive impact. Prioritizing psychological safety contributes to a humane digital ecosystem and strengthens the foundation for inclusive and sustainable AI development.





3. Economic/Property



What is this?

Economic or property harm refers to financial loss or damage to assets resulting from the use or misuse of AI systems. This includes wrongful denial of credit, job displacement due to automation, data-driven fraud, or biased decisions affecting property access and wealth accumulation. Consumers, workers, businesses, and governments may suffer from these harms through reduced income, lost opportunities, or compromised economic stability. AI's capacity to manage financial decisions, conduct transactions, or assess eligibility means that errors or biases can lead to significant economic consequences. Addressing such harm requires robust legal protections, algorithmic transparency, and fair decision-making processes.

How is it relevant?

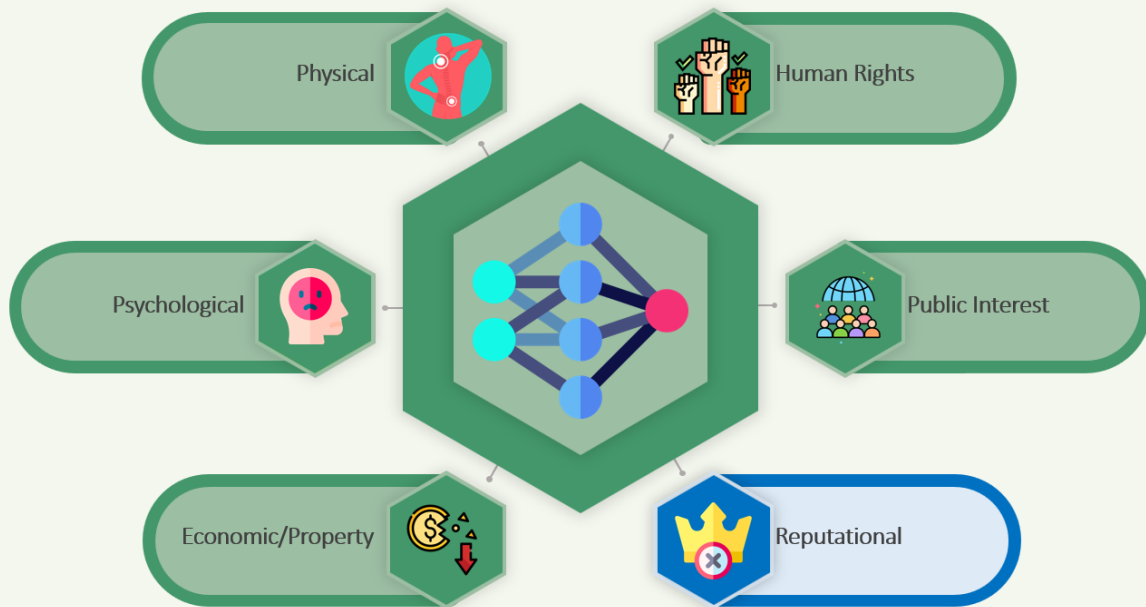
As AI systems increasingly influence economic decisions—such as credit scoring, hiring, and pricing—they shape individuals' and organizations' financial prospects. Faulty or biased algorithms can lead to misallocations of resources, deepening inequality and limiting social mobility. Businesses may suffer operational losses due to AI errors or face litigation over unfair practices. Governments managing public services with AI risk misusing funds or undermining social safety nets. Ensuring economic fairness and resilience requires evaluating how AI systems impact wealth distribution, employment dynamics, and market competition. Thus, economic harm is central to discussions of AI ethics, governance, and accountability.

Why does it matter?

Unchecked economic harm from AI systems can widen the wealth gap, reduce access to essential resources, and destabilize markets. It undermines the principle of fairness, especially if disadvantaged groups face greater financial risks. On a macro level, economic harm can stifle innovation by concentrating power in the hands of a few dominant players. Public disillusionment may follow if AI systems appear to perpetuate or exacerbate inequality. Protecting against economic harm strengthens social trust, economic inclusion, and the equitable benefits of technological progress. As economies digitize, ethical AI governance becomes key to ensuring that innovation serves the many, not just the few.



4. Reputational



What is this?

Reputational harm refers to the damage done to a company or entity's public image or trustworthiness as a result of AI-related incidents, such as biased algorithms, system malfunctions, or unethical use. A company's reputation can be severely impacted if its AI system causes harm or is seen as unreliable, untrustworthy, or discriminatory. This harm can result from negative media coverage, customer dissatisfaction, regulatory scrutiny, or public backlash. The long-term effects of reputational damage may include loss of customer loyalty, decreased sales, and difficulty in attracting partners or investors.

How is it relevant?

Reputational harm is highly relevant in an AI-driven environment because trust is essential for both consumers and businesses. In today's interconnected world, negative information can spread rapidly through social media and news outlets, potentially leading to widespread damage. Companies that deploy AI systems have to prioritize ethical AI practices, transparency, and accountability. If an AI failure harms consumers or employees, it may result in significant reputational damage, impacting future business prospects. In sectors like healthcare, finance, and public safety, where trust is paramount, AI failures or perceived misuse can permanently tarnish a company's public image.

Why does it matter?

Reputational damage can have lasting consequences for businesses, making it more difficult to recover from AI-related incidents. Customers, once dissatisfied or harmed by a product, may choose to go elsewhere, resulting in a loss of revenue and market share. Furthermore, negative public perception can lead to increased regulatory scrutiny and legal repercussions. Trust is a key pillar for maintaining a successful business, and reputational harm can affect every aspect, from sales to partnerships, employee morale, and customer loyalty. Therefore, businesses must take proactive measures to mitigate risks associated with AI deployments to protect their reputation and ensure long-term success.



5. Public Interest



What is this?

Public interest harm refers to the adverse impact that AI systems can have on the general population, communities, or society at large. This includes the potential for AI systems to cause societal disruptions, undermine democratic processes, spread misinformation, or exacerbate inequality. Examples include the use of AI in political manipulation, deepfakes, or biased law enforcement practices that disproportionately affect certain communities. Public interest harm can also include the broader societal implications of job displacement, privacy erosion, or the increased surveillance of citizens. Such harm affects the collective well-being and societal trust in technology.

How is it relevant?

Public interest harm is highly relevant because AI systems are increasingly integrated into aspects of everyday life—healthcare, transportation, education, and law enforcement. The failure to address potential risks in these areas can undermine societal stability and trust in institutions. Governments, regulators, and companies must ensure that AI technologies are deployed in ways that prioritize the collective good and protect vulnerable populations. As AI systems become more ubiquitous, their impact on the public interest will grow, making it essential to balance innovation with safeguards to protect societal welfare and prevent harm.

Why does it matter?

Public interest harm matters because it can lead to widespread societal consequences. AI systems that negatively affect large groups of people can reduce public trust in the technology, hinder social progress, and create long-term systemic problems. If AI systems worsen inequality, infringe upon civil liberties, or fuel misinformation, they can destabilize social structures and damage democratic values. Therefore, addressing public interest harm is critical to ensuring that AI contributes positively to society and does not erode public trust in technology or institutions. Governments and businesses must prioritize responsible AI practices that align with the greater good.



6. Human Rights



What is this?

Human rights harm refers to the violation or infringement of fundamental human rights due to the implementation or operation of AI systems. This could include AI-driven surveillance that violates privacy rights, discriminatory algorithms that deny equal access to services, or AI systems that disproportionately harm marginalized groups. Other examples include AI used to infringe upon freedom of expression, unjustly penalizing individuals based on biased data, or violating labor rights through AI-powered automation without providing adequate worker protections. Ensuring that AI respects and promotes human rights is essential for creating an ethical, inclusive, and just society.

How is it relevant?

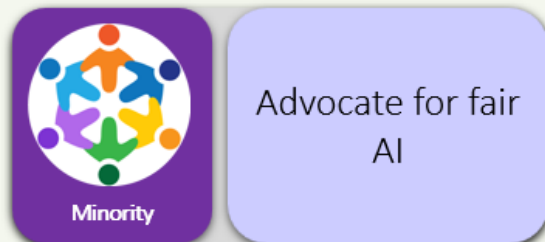
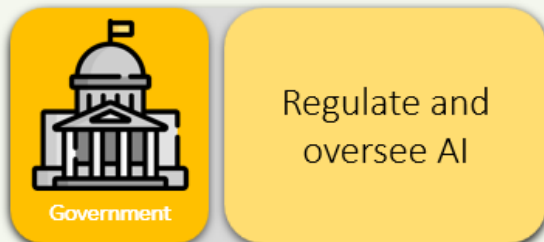
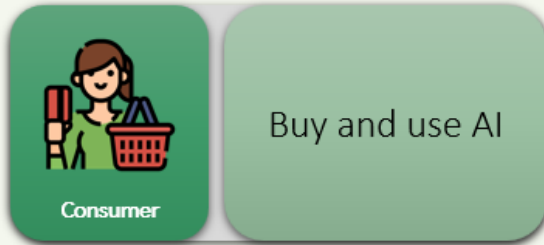
Human rights harm is relevant in the context of AI because the technologies used in AI can impact the core freedoms and rights of individuals. The ability of AI systems to gather, analyze, and make decisions about individuals' data brings new challenges for the protection of rights such as privacy, freedom of speech, and non-discrimination. In a global context where AI is being used for a range of purposes, it is crucial to ensure that AI systems do not inadvertently or intentionally violate human rights, especially in sensitive areas like law enforcement, employment, and healthcare.

Why does it matter?

Human rights harm matters because it challenges the ethical foundations of AI development and implementation. If AI systems violate basic human rights, it can result in serious consequences for individuals, communities, and societies. For example, data misuse or discriminatory algorithms can perpetuate inequality and injustice, harming the dignity and autonomy of affected individuals. Moreover, the violation of human rights can lead to legal challenges, reputational damage, and loss of public trust in AI. Ensuring that AI respects and promotes human rights is essential for fostering trust, social cohesion, and the responsible use of technology in a democratic society.



Mapping Conformity Assessments to ecosystem actors





Serious Incident Reporting		Actor					
Stage	Description	Consumer	Worker	Business	Government	Minority	General Public
Physical	Involves injury, illness, or death caused by an AI system's malfunction or decision-making. This may result from incorrect diagnostics, faulty autonomous operations, or harmful recommendations. High-risk AI used in healthcare, transportation, or machinery control could pose serious physical dangers to individuals or groups if not properly monitored or contained.	Consumers may be harmed physically by malfunctioning AI products, such as faulty autonomous vehicles, misdiagnosing medical devices, or poorly designed smart home systems. These failures can lead to accidents, injuries, or even fatalities. Such events can significantly undermine public trust in AI, discouraging adoption of beneficial technologies. Furthermore, consumers may face challenges pursuing justice or compensation due to complex liability issues involving AI. Vulnerable populations, such as the elderly or	AI systems deployed in workplaces, such as robotic arms or autonomous vehicles in warehouses and factories, can pose serious safety risks if not properly managed. Faulty sensors, flawed decision-making algorithms, or integration errors may result in collisions, crushing injuries, or other physical incidents. Beyond direct accidents, poorly implemented AI can lead to unsafe work environments due to system unpredictability or lack of proper human-machine coordination. Workers may also be denied protective	Businesses utilizing AI systems that cause physical harm face significant consequences, including legal liability, compensation claims, damaged reputations, and disrupted operations. Insurance costs may rise, and relationships with customers, partners, and regulators can be strained. Physical harm incidents also invite intense scrutiny from regulators and the public, potentially leading to operational restrictions or market withdrawals. Internally,	Governments must respond swiftly and effectively to physical harm caused by AI systems to protect public safety. This includes enforcing existing safety laws, developing AI-specific regulations, and overseeing compliance through market surveillance authorities. Serious physical incidents can provoke public fear, reduce confidence in emerging technologies, and result in political pressure for stricter controls. Governments must also coordinate with	Minorities may experience disproportionate physical harm from AI systems due to underrepresentation in training data and a lack of inclusive design. For example, facial recognition systems may fail to identify darker skin tones accurately, leading to misidentification and physical risks in high-stakes environments like policing or border control. Automated healthcare tools may not account for diverse physiological traits, resulting in misdiagnoses. Additionally, minorities may reside in areas where lower-quality or experimental AI systems are deployed first, increasing exposure	Physical harm incidents involving AI systems can have wide-reaching implications for the general public. Such events may undermine collective trust in emerging technologies, stall innovation, and amplify calls for restrictive regulation. Infrastructure-related AI—such as traffic management or emergency response systems—if flawed, can result in large-scale harm affecting entire communities. Schools, transportation, healthcare, and other public



		disabled, may be especially at risk if products are not designed inclusively. Ongoing safety assurance, transparent reporting, and responsive support systems are essential to protect consumers from physical AI-related harm.	interventions if harm is not promptly reported or investigated. Effective worker training, comprehensive AI risk assessments, and strong occupational health and safety regulations are crucial to reducing these dangers.	incidents may create low employee morale and increase turnover. Businesses must therefore prioritize safety by adopting rigorous testing protocols, incident reporting mechanisms, and compliance with AI-specific safety standards. A proactive approach not only protects users but also secures long-term business sustainability and trust.	international regulatory bodies to standardize safety requirements. Additionally, public health and emergency services may be strained in responding to incidents. Investing in robust regulatory frameworks, response protocols, and continuous monitoring ensures government readiness to handle the risks of AI-induced physical harm.	to risk. Addressing these disparities requires equitable data practices, inclusive testing, and targeted safety evaluations to ensure AI systems protect all demographics equally.	services may also become unreliable if AI systems perform dangerously. Additionally, widespread harm may necessitate public expenditure for remediation, emergency response, and legal adjudication. Ensuring public safety through clear accountability mechanisms, transparent incident reporting, and preventive AI design is vital for societal resilience.
Psychological	Refers to mental distress, trauma, or emotional damage stemming from AI interactions or outputs. This could include manipulative	AI systems can contribute to psychological harm for consumers through manipulation, over-surveillance, or biased decision-making.	Psychological harm to workers can arise when AI systems monitor performance excessively, automate evaluations, or replace human	Psychological harm among employees or consumers linked to AI use can pose reputational and operational risks for businesses. A stressed or	Governments have a duty to prevent psychological harm from AI systems by establishing legal safeguards and ethical	Minorities may be disproportionately affected by psychological harm from AI due to biased algorithms, cultural insensitivity, or surveillance-focused	The general public may experience widespread psychological harm due to the societal effects of pervasive AI use—such as



	content, discriminatory profiling, or biased decision-making affecting mental well-being. Psychological harm often results in anxiety, loss of dignity, or psychological disorders, especially in vulnerable individuals exposed to high-risk or immersive AI systems.	Recommendation algorithms may reinforce negative behaviors, fuel anxiety, or promote harmful content. If AI-driven systems make consumers feel monitored or judged, they may experience stress or diminished self-worth. Biased outputs—such as unfair loan denials or criminal risk assessments—can trigger feelings of helplessness and injustice. Moreover, consumers might suffer from uncertainty or fear regarding how AI systems operate, especially if they lack transparency. Ensuring that AI systems are explainable, fair, and respectful of user autonomy is essential to safeguarding	decision-making in sensitive areas like promotions or disciplinary actions. Such practices may foster stress, anxiety, and burnout, especially when systems are opaque or perceived as unfair. Job insecurity due to AI-driven automation can also create ongoing psychological strain. The feeling of being constantly observed or judged by an algorithm can erode morale and reduce job satisfaction. To protect mental health, organizations must design AI tools that are transparent, support well-being, and integrate human	demoralized workforce may be less productive, less creative, and more likely to experience turnover. Negative media coverage or public outcry around AI-induced anxiety—such as bias in hiring tools or toxic content algorithms—can damage brand trust. Companies must be proactive in assessing and mitigating psychological risks through impact assessments, ethical design, and stakeholder engagement. Transparent communication about how AI systems work and active participation in mental health initiatives	frameworks. Psychological distress caused by algorithmic decisions can lead to public dissatisfaction, protests, and reduced trust in institutions. Governments must ensure that AI deployments in public services—such as welfare or policing—do not humiliate or stigmatize users. Regulatory bodies should mandate mental health impact assessments and promote transparency. In partnership with health agencies and experts, governments can set standards for humane, psychologically safe AI. These efforts enhance social cohesion,	applications. Predictive policing or immigration AI tools may foster feelings of fear, discrimination, and mistrust among targeted communities. Systems that fail to recognize non-majority linguistic or social norms can cause alienation or distress. Minority users may also face microaggressions through biased chatbot interactions or unfair recommendations. Such harms may not be visible but can have deep impacts on mental health and community cohesion. Inclusive AI development, robust feedback mechanisms, and community involvement are key to preventing psychological harm and promoting	information overload, disinformation, and increased digital dependency. Constant exposure to algorithmic content can foster echo chambers, social comparison, or anxiety. Trust in digital systems may erode when decisions lack transparency or appear unjust. Communities affected by biased or malfunctioning AI systems may feel disempowered or marginalized. Public education on digital literacy, mental health resilience, and responsible AI development is crucial. Ensuring algorithmic transparency, setting ethical
--	---	--	--	---	--	--	---





		mental well-being and building public trust in technology.	oversight, while maintaining clear communication about the role and limitations of AI.	demonstrate corporate responsibility and contribute to long-term organizational resilience and credibility.	protect vulnerable populations, and help integrate technology into society in a balanced, ethical way.	equitable experiences.	boundaries for AI influence, and promoting mental well-being across populations helps safeguard the psychological fabric of society.
Economic/Property	Includes financial loss, economic disadvantage, or damage to assets caused by AI system errors or misuse. Examples include flawed credit scoring, unjust job rejection, or algorithmic trading failures. Such harm may affect individuals, businesses, or institutions and often results from systemic bias, technical defects, or unauthorized AI-driven transactions.	AI systems that incorrectly assess creditworthiness, mismanage personal financial data, or enable fraud can cause direct economic harm to consumers. These issues may result in wrongful denial of loans, unauthorized transactions, or exploitation through dynamic pricing algorithms. Errors in AI-driven insurance assessments or job application screeners can also reduce economic opportunities. Consumers may struggle to contest	Workers are particularly vulnerable to economic harm from AI systems that automate tasks or replace entire roles, leading to job displacement or wage suppression. Inaccurate performance monitoring or biased algorithmic evaluations can result in wrongful termination or missed promotions. Gig workers, often managed by opaque AI platforms, may experience unpredictable income and unfair	Economic harm for businesses can stem from AI errors that disrupt operations, create legal liabilities, or alienate customers. Malfunctioning AI in logistics, finance, or customer service may lead to loss of revenue, reputational damage, and costly litigation. Intellectual property theft or data breaches facilitated by AI misuse can undermine competitiveness. Additionally, poor	Governments can face economic harm when public AI systems misallocate resources, mismanage data, or fail to prevent fraud. Budget overruns, legal battles, and public backlash may follow poorly implemented AI projects. Misjudgments in AI-driven social welfare, taxation, or procurement tools can lead to inefficiencies and loss of public funds. Additionally, governments bear the cost of	Minorities may suffer economic harm when AI systems perpetuate existing biases in areas like lending, employment, or housing. Discriminatory algorithms can result in lower approval rates for loans or credit, biased job screening, and exclusion from economic opportunities. These effects compound historic inequalities, limiting upward mobility. Additionally, marginalized communities may be more exposed to predatory AI-driven	Economic harm to the general public from AI arises when systemic flaws lead to market disruptions, increased inequality, or financial instability. For instance, algorithmic trading errors can crash stock markets, while biased AI in employment or credit access can widen socioeconomic divides. Misuse of AI in public services can also erode trust and effectiveness, straining



		such outcomes due to opaque decision-making processes and lack of accountability. This economic insecurity can deepen inequalities, especially for those already vulnerable. Transparent AI models, fair data practices, and accessible dispute resolution mechanisms are essential to protect consumers from economic harm and foster financial autonomy.	deactivations. Furthermore, workers may lack legal protections or bargaining power against AI-driven economic impacts. These harms can widen the income gap and reduce social mobility. To counter this, AI regulation must emphasize fair labor practices, retraining programs, and algorithmic transparency to ensure workers are treated justly and economically secure.	AI governance may result in regulatory fines or the withdrawal of market access. Small and medium enterprises may struggle to adapt, creating economic disparities. Businesses must invest in AI oversight, ethical implementation, and contingency planning. Done right, this mitigates economic harm while enhancing trust, innovation, and market stability.	economic harm to citizens, including unemployment benefits and legal remediation. To prevent this, public sector AI use must be transparent, accountable, and aligned with social welfare goals. Investing in regulatory capacity, digital infrastructure, and public sector AI literacy supports long-term fiscal sustainability and public trust.	services or suffer from digital redlining. Without transparency or accessible redress mechanisms, these harms often go unchallenged. It is crucial to design AI systems with fairness, inclusivity, and representative data in mind, ensuring all groups can benefit from and participate in the digital economy.	communal resources. Public confidence in the digital economy may falter if AI is seen as causing more harm than good. Proactive governance, public education, and equitable AI policy are essential to distribute economic benefits fairly and safeguard the prosperity of society as a whole.
Reputational	Occurs when an AI system's output damages an individual's or organization's public image or credibility. This can stem from false accusations, inaccurate content generation, or	Consumers may suffer reputational harm when AI systems mishandle or expose personal information, such as identity data, financial history, or online behavior. Incorrect or biased outputs—like false	AI systems that evaluate, monitor, or communicate about workers can significantly impact their reputations. Errors in AI-driven performance assessments or surveillance tools	Businesses risk reputational damage if their AI systems cause harm or controversy—such as biased algorithms, discriminatory outputs, or major technical failures.	Governments deploying AI systems in public services can face reputational harm when those systems cause injustice, bias, or failures that affect citizens. Missteps in areas	Reputational harm to minorities often stems from AI systems that mislabel, misrepresent, or stigmatize individuals based on biased data. For example, AI used in law enforcement or	Widespread reputational harm caused by AI can lead to broader social consequences, including reduced trust in technology, polarized discourse, and



	flawed moderation systems. Reputational harm may be irreversible, affecting personal or professional standing, career prospects, or business relationships—especially when amplified by social or digital media.	criminal profiling or social media moderation errors—can damage a person’s reputation publicly or within institutions. These effects may spread rapidly online, making recovery difficult. Even being flagged by an opaque AI system can stigmatize individuals, impacting employment, relationships, or community standing. Vulnerable consumers often lack the means to correct these records or seek redress. Privacy-focused design, explainable outputs, and consumer empowerment are crucial for protecting reputations and	may lead to unfair reviews, missed promotions, or even job termination. Automated HR tools might share inaccurate data across platforms or flag workers in ways that stigmatize them professionally. These reputational issues can hinder future job prospects and erode trust in employers. For gig workers or freelancers, platform ratings shaped by AI can be critical to income. Organizations must ensure their AI systems are fair, accurate, and provide opportunities for human oversight and correction to protect worker reputations.	Public perception can shift rapidly, especially in high-profile incidents involving customer mistreatment or ethical concerns. Negative media coverage, regulatory fines, and social media backlash can diminish brand value, investor confidence, and customer loyalty. B2B relationships may also suffer if trust in a company’s AI solutions declines. Reputation is especially critical in sectors like healthcare, finance, and education. Proactive risk management, ethical design, and transparent communication about AI practices help businesses	like facial recognition, predictive policing, or welfare distribution can trigger public backlash and decrease trust in institutions. Transparency failures or lack of recourse mechanisms amplify these effects. Once trust is eroded, it can become difficult to pass future digital or regulatory reforms. Government reputation also affects international credibility and diplomatic relations, especially in cross-border data or AI safety cooperation. Responsible AI use, strong oversight, and	employment may disproportionately flag minority individuals, impacting how they’re perceived by institutions or peers. Cultural misunderstandings embedded in AI can reinforce stereotypes or invisibilize diverse identities. These harms can result in social exclusion or limited access to opportunities. Because reputational damage is often widespread and hard to reverse, minorities may face long-term disadvantages. Inclusive data practices, culturally competent design, and accountability mechanisms are essential to prevent reputation-based discrimination and uphold dignity for all.	challenges to social cohesion. When AI systems spread false information, enable deepfakes, or amplify harmful content, they may damage the reputations of public figures, private citizens, or institutions. Such harms can erode civil discourse, misinform communities, and destabilize democratic processes. Public sentiment may shift toward skepticism of AI’s role in society. Preventing reputational harm at scale requires coordinated efforts in digital literacy, content moderation, misinformation control, and
--	---	--	---	--	---	--	--



		ensuring fair AI-driven decisions.		safeguard reputation and build long-term public and stakeholder trust.	engagement with public concerns are vital for maintaining institutional legitimacy.		transparent governance of AI technologies across sectors and platforms.
Public Interest	Encompasses threats to societal stability, democratic processes, or public safety due to high-risk AI deployment. This includes algorithmic misinformation, electoral manipulation, or undermining public health strategies. Harm to public interest often affects large populations and requires urgent intervention, especially when the AI system's influence disrupts social cohesion or governance.	Consumers may be affected by AI systems that undermine public interest, such as those that spread misinformation, violate privacy at scale, or compromise the integrity of democratic processes. If AI-driven platforms manipulate political discourse or suppress truthful content, consumers may unknowingly form biased opinions or lose trust in public institutions. Public interest is also harmed when AI exacerbates inequality, environmental damage, or safety risks. Since	Workers can be adversely impacted when AI systems are deployed in ways that conflict with the public interest—such as replacing human labor without social protections, reinforcing unsafe labor practices, or weakening unions. AI-driven efficiencies may increase productivity but contribute to broader job polarization or regional economic decline. Public interest is compromised when technological progress benefits a few at the expense of working-class	Businesses depend on societal stability and consumer trust, both of which suffer when AI systems threaten public interest. For instance, companies that deploy or are associated with harmful AI—such as tools that erode privacy, spread misinformation, or disrupt civic systems—risk backlash and long-term reputational damage. These harms also invite regulatory crackdowns that affect the entire sector. While innovation is vital,	Governments bear the responsibility for ensuring that AI systems operate in ways that uphold the public interest. This includes regulating AI that influences democratic discourse, national security, or critical infrastructure. When AI applications violate public trust—through surveillance overreach, biased decision-making, or lack of transparency—citizen confidence in governance erodes. The government	Minorities often experience unique vulnerabilities when AI systems fail to serve the public interest, as these failures can intensify exclusion, discrimination, or marginalization. For example, AI tools used in surveillance, policing, or social services may disproportionately target or overlook minority communities. Public interest is compromised when these groups are systematically disadvantaged in education, employment, or access to justice due to algorithmic bias. Additionally, minorities may be excluded from the	The general public relies on the alignment of AI with public interest to ensure social stability, democratic integrity, and collective well-being. When AI systems are used to manipulate elections, spread disinformation, or undermine public services, the entire society can suffer. These harms may not be immediately visible but can erode trust in institutions, polarize communities, and destabilize economies. Public interest harm affects everyone by





		consumer behavior and welfare are closely tied to societal stability, these broader harms can diminish confidence in markets and digital technologies. Ensuring that AI systems are aligned with collective well-being helps safeguard informed and equitable participation in society.	security and dignity. Workers also suffer when essential infrastructure—like transport or healthcare—fails due to flawed AI. Safeguarding public interest through inclusive labor policies, ethical AI design, and government accountability helps ensure technological gains are distributed fairly among all workers.	businesses have a responsibility to balance profit with public welfare. Operating in alignment with the public interest not only helps avoid social friction and legal challenges but also builds resilient markets and a loyal customer base in the digital age.	must also ensure that AI development supports shared prosperity and does not exacerbate inequality. Upholding the public interest means setting clear standards, fostering innovation responsibly, and maintaining open dialogue with stakeholders. Proactive regulation and ethical leadership help governments harness AI for societal benefit while preserving democratic values and social cohesion.	policymaking processes that shape AI governance. Centering minority voices, ensuring algorithmic fairness, and promoting inclusive design are essential to advancing the public interest and fostering a more equitable technological landscape.	diminishing transparency, fairness, and access to accurate information. A collective approach—encompassing strong regulation, ethical standards, public engagement, and educational outreach—is necessary to ensure AI strengthens the social contract rather than threatening it.
Human Rights	Refers to violations of fundamental rights such as privacy, non-discrimination, freedom of expression, or	Consumers' human rights can be endangered by AI systems that violate privacy, autonomy, or freedom of	AI can infringe on workers' rights by enabling invasive monitoring, biased decision-making in hiring or promotion, and	Businesses play a dual role regarding human rights: as potential violators and as protectors. Poorly	Governments have a fundamental duty to uphold human rights and must ensure that AI deployment	Minorities are especially vulnerable to AI-driven human rights violations due to systemic bias and underrepresentation in datasets.	The general public's human rights can be broadly affected when AI systems are used in ways that curtail civil



	access to justice. High-risk AI systems may infringe on rights through mass surveillance, biased profiling, or opaque decision-making. Such harm can undermine democratic values, disproportionately affect marginalized groups, and trigger legal liabilities under EU and international law.	expression. For example, AI-driven surveillance tools may monitor behavior without consent, infringing on the right to privacy. Biased algorithms in credit scoring, healthcare, or justice can lead to discriminatory outcomes, violating equality and due process rights. Additionally, opaque AI systems can prevent individuals from understanding or challenging harmful decisions, undermining the right to a fair remedy. Protecting consumer rights requires transparency, consent-based data practices, and regulatory oversight that ensures AI is used ethically and respects individual	denying access to due process in disciplinary actions. Automated management systems may reduce worker autonomy and limit freedom of association, especially when collective action is flagged as a threat by AI surveillance tools. Gig economy workers often face opaque algorithms that determine pay and task allocation, leaving them with limited recourse. Upholding labor rights in an AI-driven workplace demands transparency, the right to contest automated decisions, and protections for collective bargaining. Human-centered AI governance is	governed AI can lead to human rights abuses such as discrimination, surveillance, or exploitation, which can result in litigation, reputational damage, and loss of consumer trust. On the other hand, ethical AI practices aligned with human rights—like privacy-by-design or fairness auditing—can enhance brand reputation and regulatory compliance. Businesses must conduct human rights impact assessments, implement grievance mechanisms, and train staff in ethical AI use. Responsible AI practices help	within public institutions aligns with legal and moral obligations. When used in policing, immigration, or welfare, AI systems must not discriminate or deny individuals access to justice or basic services. Failure to prevent rights violations can erode democratic legitimacy and public trust. Governments must establish legal safeguards, conduct rights-based AI assessments, and ensure redress mechanisms for affected individuals. Promoting transparency, inclusive governance, and international human rights	Discriminatory outcomes in areas like policing, housing, or healthcare can reinforce marginalization and inequality. Surveillance technologies may target specific communities disproportionately, infringing on privacy and freedom of expression. Language and cultural barriers may also prevent minorities from accessing recourse when harmed. Addressing these injustices requires inclusive AI development, civil society engagement, and policies that protect marginalized voices. Human rights-centered AI governance ensures that technology empowers all people equitably and that historical inequities	liberties, suppress dissent, or automate inequality. Authoritarian uses of AI—like mass surveillance or facial recognition at protests—can chill free expression and public assembly. When these technologies become normalized, everyone’s rights are at risk. Additionally, AI decisions that affect access to healthcare, education, or justice must be transparent and contestable to uphold human dignity. A human rights-based approach to AI policy and development ensures that innovation serves the public good,
--	---	--	---	--	--	---	---





		dignity, choice, and equal treatment under the law.	critical to securing dignity, fairness, and workplace justice.	mitigate risk, promote long-term sustainability, and demonstrate a commitment to corporate social responsibility and human dignity.	standards is essential to developing AI policies that protect all citizens and ensure accountability across sectors and borders.	are not perpetuated by algorithmic systems in the digital age.	respects legal protections, and enhances rather than restricts civic freedom.
--	--	---	--	---	--	--	---





Calls to Action



Integrate Real-Time Safety Monitoring for High-Risk AI

Establish continuous safety monitoring systems for AI applications in healthcare, transport, or industrial automation. Use redundant fail-safes, human-in-the-loop oversight, and predictive analytics to detect and prevent potentially harmful malfunctions. Regularly audit and test AI behaviors in real-world conditions to ensure safe operation and immediate response in critical failure scenarios



Conduct Psychological Impact Assessments and Ethical Reviews

Before deployment, assess the psychological effects of AI systems, particularly those involving personalization, profiling, or social interaction. Implement content moderation, consent mechanisms, and exposure thresholds to protect users from distress. Include diverse stakeholders—psychologists, ethicists, and advocacy groups—in review boards to anticipate and mitigate mental health risks.



Implement Fairness Audits and Explainability Tools in Decision Systems

Deploy algorithmic fairness audits, particularly in systems affecting financial services, hiring, and asset management. Integrate explainability features so affected individuals can understand decisions and contest errors. Use robust validation datasets to reduce systemic bias and protect users and organizations from economic or legal fallout due to flawed AI outcomes.



Establish AI Output Validation and Crisis Response Protocols

Apply strict quality controls on generative or decision-making AI that affects public communication, moderation, or reputation. Pre-screen outputs for defamatory content, false claims, or bias. Develop a rapid response plan for correcting reputational damage, including automated takedowns, public disclosures, and legal follow-up if AI-generated content causes reputational harm



Conclusion

The introduction of mandatory serious incident reporting under Article 73 of the EU AI Act marks a pivotal evolution in the post-market governance of high-risk AI systems—ushering in a more responsive, transparent, and enforceable model of accountability. By obligating providers and, where relevant, deployers to report incidents that pose risks to health, safety, or fundamental rights, Article 73 translates the Act’s overarching commitment to human-centric AI into a concrete operational safeguard. This reporting mechanism reduces regulatory blind spots, enhances system oversight, and supports the continuous alignment of AI applications with EU legal and ethical standards.

Throughout this paper, we have demonstrated how Article 73 serves as a critical point of convergence for AI developers, users, regulators, and other stakeholders—fostering a shared responsibility for detecting, communicating, and mitigating serious incidents. When embedded thoughtfully into the AI lifecycle, incident reporting contributes not only to regulatory compliance, but also to the ethical and operational robustness of AI systems in the real world.

We examined each phase of AI deployment—from initial design choices to post-deployment monitoring and eventual system retirement—to show that serious incident reporting should not be treated as a reactive legal obligation but as an integral component of risk-informed innovation. When properly integrated, it helps embed traceability, transparency, and human oversight into the operational DNA of AI systems.

Yet, effective implementation of Article 73 demands more than formal reporting channels. It requires organizational preparedness, clear internal protocols, and a culture of accountability that prioritizes early detection and open communication. This can be particularly challenging for smaller firms or new entrants to the market, making early planning, capacity building, and structured engagement with regulators indispensable.

Despite these challenges, forward-looking companies, supervisory authorities, and civil society actors are already demonstrating that robust incident reporting can mitigate harm, enhance system reliability, and reinforce public confidence. As AI systems become increasingly embedded in sensitive domains—such as healthcare, education, employment, and law enforcement—Article 73 plays a vital role in ensuring that AI remains lawful, trustworthy, and aligned with European democratic values.





References

Future of Life Institute, (2025), 'The EU Artificial Intelligence Act', accessible at <https://artificialintelligenceact.eu/> (last accessed 7th May 2025)

European Parliament and The Council of the European Union, (2024), 2024/1689 Regulation (EU) 2024/1689 of the European Parliament and of The Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act), accessible at https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=OJ:L_202401689 (last accessed 7th May 2025)

OECD AI, (2025), 'AIM: AI Incidents and Hazards Monitor', accessible at: https://oecd.ai/en/incidents?search_terms=%5B%5D&and_condition=false&from_date=2014-01-01&to_date=2025-10-10&properties_config=%7B%22principles%22:%5B%5D,%22industries%22:%5B%5D,%22harm_types%22:%5B%5D,%22harm_levels%22:%5B%5D,%22harmed_entities%22:%5B%5D,%22business_functions%22:%5B%5D,%22ai_tasks%22:%5B%5D,%22autonomy_levels%22:%5B%5D%7D&order_by=date&num_results=20 (last accessed 10th October 2025)





Important notice

This document has been prepared by AI & Partners B.V. for the sole purpose of enabling the parties to whom it is addressed to evaluate the capabilities of AI & Partners B.V. to supply the proposed services.

Other than as stated below, this document and its contents are confidential and prepared solely for your information, and may not be reproduced, redistributed or passed on to any other person in whole or in part. If this document contains details of an arrangement that could result in a tax or National Insurance saving, no such conditions of confidentiality apply to the details of that arrangement (for example, for the purpose of discussion with tax authorities). No other party is entitled to rely on this document for any purpose whatsoever and we accept no liability to any other party who is shown or obtains access to this document.

This document is not an offer and is not intended to be contractually binding. Should this proposal be acceptable to you, and following the conclusion of our internal acceptance procedures, we would be pleased to discuss terms and conditions with you prior to our appointment. Images used throughout the document have either been produced in-house or sourced from publicly available sources (see **References** for details).

AI & Partners B.V. is the Dutch headquarters of AI & Partners, a global professional services firm. Please see <https://www.ai-and-partners.com/> to learn more about us.

© 2025 AI & Partners B.V. All rights reserved.